

6(a)	an arrow through phospholipid layer in outer membrane <u>and</u> an arrow through phospholipid layer in inner membrane ; R arrows through proteins or A, B, C or D	1
6(b)	1 D = ATP synthase / ATP synthetase ; R ATPase 2 A, B and C = ETC / electron transport chain / electron carriers / electron acceptors ; <i>plus any five from:</i> <i>role of, ETC proteins / A, B and C</i> 3 electrons travel along ETC ; 4 energy released ; 5 (use energy released) to pump / actively transport, H^+ / protons, into intermembrane space ; 6 increases concentration of H^+ / protons (in intermembrane space) / creates proton gradient ; <i>role of D</i> 7 H^+ / protons, diffuse through, ATP synthase / D ; 8 (from intermembrane space) to matrix ; 9 synthesis of ATP from ADP and P_i ; 10 chemiosmosis ;	7

Question	Answer	Marks
6(c)	any three from: 1 (named) enzyme for, Krebs cycle / link reaction ; 2 ETC <u>proteins</u> / cytochromes / electron carriers / electron acceptors / A / B / C ; 3 ATP synthase / D / described ; 4 membrane / channel / transport / carrier / structural, proteins ; 5 (make) tRNA / rRNA / ribosomes ;	3

Question	Answer	Marks
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